You have invented a new form of the game of darts, and it is based on the idea that the tightest cluster of darts wins — but they don't necessarily have to be close to a target. In fact, the game is easy to play because it doesn't require a target; players can just throw darts against a wall (preferably one that you don't care about). When a player is done throwing their n darts, they wrap a string tightly around the outside of all the outermost darts, so that the string encompasses all of the thrown darts. If s is the length of the string needed to enclose the darts, then the score for that player's darts is $100 \cdot n/(1+s)$.

Now you need a program to determine the score of different dart configurations.

Input

Input consists of up to 200 player turns, one turn per line. Each player's turn has a list of 1 to 100 pairs of real numbers with at most 3 digits past the decimal, which are the (x, y) locations where the dart has landed. The bounds are $-30 \le x, y \le 30$. Input ends at the end of file.

Output

For each turn, print the score obtained, with at most 0.0001 error.

Sample Input 1

```
10.8 -13.7
0.278 2.555 2.815 3.800 3.920 1
2.358 1.731 0.663 3.485 4.276 6
0.117 5.881 4.655 2.766 0.941 0
```

Sample Output 1

```
100.0000000000
29.5344250128
34.3557840779
30.3425374145
```